Problem 7

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Problem 1 *Find values for a, b, and c so that the following function is differentiable everywhere.*

$$f(x) = \begin{cases} a\sin(x) + b\cos(x) & \text{if } x < 0 \\ ax^2 + bx + c & \text{if } x \geq 0 \end{cases}$$

Explanation. The function f is differentiable for $x < 0$ since f is a combination of trigonometric functions. f is also differentiable $x > 0$ since f is a polynomial on that interval. We need to focus on $x = 0$. Since f is differentiable everywhere, f must also be differentiable at $x = 0$ and therefore continuous at $x = 0$. Therefore:

We need that $\lim\limits_{x\to 0^-} f(x) = \lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0} f(x) = f(0)$. Observe that

- $\lim\limits_{x\to 0^-} f(x) = \lim\limits_{x\to 0^-} (a\sin(x) + b\cos(x)) = b(1) = b.$
- $\lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0^+} (ax^2 + bx + c) = c.$

Thus, we must have that $b = c$

Next, if $f'(0)$ exists, then:

$$f'(0) = \lim_{x\to 0} \frac{f(x) - f(0)}{x - 0}$$

$$\text{since } f(0) = a \cdot 0^2 + b \cdot 0 + c = c$$

$$\lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0} \frac{f(x) - c}{x}$$

Since this limit exists, the left and right limits must be equal.

$$\begin{aligned}
\lim_{x\to 0^+} \frac{f(x)-c}{x} &= \lim_{x\to 0^+} \frac{ax^2+bx+c-c}{x} \\
&= \lim_{x\to 0^+} \frac{ax^2+bx}{x} \\
&= \lim_{x\to 0^+} (ax+b) \\
&= b
\end{aligned}$$

$$\begin{aligned}
\lim_{x\to 0^-} \frac{f(x)-c}{x} &= \lim_{x\to 0^-} \frac{a\sin(x)+b\cos(x)-c}{x} \\
&= \lim_{x\to 0^-} \frac{a\sin(x)}{x} + \lim_{x\to 0^-} \frac{b\cos(x)-c}{x}
\end{aligned}$$

we already found $b = c$ so we have

$$\lim_{x\to 0^-} \frac{a\sin(x)}{x} + \lim_{x\to 0^-} \frac{c\cdot\cos(x)-c}{x}$$

$$= a\cdot \lim_{x\to 0^-} \frac{\sin(x)}{x} + c\cdot \lim_{x\to 0^-} \frac{\cos(x)-1}{x}$$

since $\lim_{x\to 0} \frac{\sin(x)}{x} = 1$ and $\lim_{x\to 0} \frac{\cos(x)-1}{x} = 0$ we have

$$a\cdot \lim_{x\to 0^-} \frac{\sin(x)}{x} + c\cdot \lim_{x\to 0^-} \frac{\cos(x)-1}{x} = a$$

This means in order for the derivative to exist, $a = b = c$.